MODEL LAB · 2026-10-28

# Every prior, same slate

The held-out scores that decided which prior is served, then how each one would have ranked today's hitters.

| Served (full season) | 11 | 17,800 | 2.92% | 0.5998 |
|---|---|---|---|---|
| served prior | priors compared | held-out plate appearances | held-out HR rate | best AUC · KNN comps |

## Held-out scores

Fitted on plate appearances before 2026-09-13, scored on the 17,800 that came after (actual HR rate 2.92%). No outcome in the test window was visible during fitting. **Read AUC first** — the slate ranks hitters and takes the top six, so ordering is what matters. Brier and log loss are dominated by the ~3% base rate and barely separate the priors. Metrics are per plate appearance; the probabilities lower down are per game.

| PRIOR | AUC | TOP-DECILE LIFT | BRIER | LOG LOSS | CAL. RMSE |
|---|---|---|---|---|---|
| **Flat league**<br>League HR rate as the prior for every hitter. | 0.5813 | 1.62× | 0.02826 | 0.13096 | 0.00454 |
| **KNN comps**<br>Prior is the pooled rate of the nearest comparables. | 0.5998 | 1.64× | 0.02821 | 0.12987 | 0.00605 |
| **SVR**<br>Support-vector regression on the same feature matrix. | 0.5925 | 1.52× | 0.02826 | 0.13049 | 0.00693 |
| **Random forest**<br>Forest regression; a learned similarity metric. | 0.5951 | 1.50× | 0.02825 | 0.13018 | 0.00678 |
| **Core (KNN+SVR)**<br>Blend of KNN and SVR, no forest. | 0.5998 | 1.64× | 0.02821 | 0.12987 | 0.00605 |
| **Forest blend**<br>KNN + SVR + forest, weights fitted out-of-fold. | 0.5998 | 1.64× | 0.02821 | 0.12987 | 0.00605 |
| **Linear regression**<br>Ordinary least squares on the same features. | 0.5964 | 1.58× | 0.02825 | 0.13026 | 0.00487 |
| **Logistic regression**<br>Binomial logistic on the per-PA home run outcome. | 0.5953 | 1.60× | 0.02830 | 0.13055 | 0.00609 |
| **Neural net (torch)**<br>Two-layer PyTorch MLP, PA-weighted, trained on the same features. | 0.5955 | 1.56× | 0.02824 | 0.13016 | 0.00503 |
| **XGBoost**<br>Gradient-boosted trees, depth 3 with subsampling and L2. | 0.5945 | 1.45× | 0.02829 | 0.13055 | 0.00733 |

### BLEND WEIGHTS (FOREST VARIANT)

KNN **1.00** SVR **0.00** RF **0.00**

### SHRINKAGE K PER PRIOR

league **260** knn **260** svr **260** rf **260** core **260** ensemble **260** linear **260** logistic **260** nn **260** xgb **260**

Estimated by method of moments on out-of-fold residuals, not hand-tuned.

## Thin-sample subset

The same held-out window, restricted to hitters with fewer than 100 plate appearances before the cutoff (1,620 PA, actual HR rate 2.35%). For a regular with 600 PA the observed record swamps any prior, so the priors mostly separate here.

| PRIOR | AUC | TOP-DECILE LIFT | BRIER | LOG LOSS | CAL. RMSE |
|---|---|---|---|---|---|
| **Flat league**<br>League HR rate as the prior for every hitter. | 0.6456 | 1.58× | 0.02291 | 0.11126 | 0.01790 |
| **KNN comps**<br>Prior is the pooled rate of the nearest comparables. | 0.6062 | 1.58× | 0.02288 | 0.10980 | 0.01007 |
| **SVR**<br>Support-vector regression on the same feature matrix. | 0.5783 | 1.58× | 0.02323 | 0.11613 | 0.02309 |
| **Random forest**<br>Forest regression; a learned similarity metric. | 0.6000 | 1.32× | 0.02320 | 0.11319 | 0.02260 |
| **Core (KNN+SVR)**<br>Blend of KNN and SVR, no forest. | 0.6062 | 1.58× | 0.02288 | 0.10980 | 0.01007 |
| **Forest blend**<br>KNN + SVR + forest, weights fitted out-of-fold. | 0.6062 | 1.58× | 0.02288 | 0.10980 | 0.01007 |
| **Linear regression**<br>Ordinary least squares on the same features. | 0.6112 | 1.32× | 0.02322 | 0.11383 | 0.02025 |
| **Logistic regression**<br>Binomial logistic on the per-PA home run outcome. | 0.6015 | 1.32× | 0.02354 | 0.11442 | 0.02384 |
| **Neural net (torch)**<br>Two-layer PyTorch MLP, PA-weighted, trained on the same features. | 0.6056 | 1.58× | 0.02300 | 0.11128 | 0.01557 |
| **XGBoost**<br>Gradient-boosted trees, depth 3 with subsampling and L2. | 0.5934 | 1.05× | 0.02361 | 0.11640 | 0.02762 |

## Slate agreement

| PRIOR | SHARED | ITS TOP SIX |
|---|---|---|
| **Served (full season)** | 6/6 | Jackson Chourio 15.4%, Junior Caminero 14.9%, Jake Bauers 14.1%, William Contreras 13.8%, Jonathan Aranda 11.7%, Christian Yelich 11.6% |
| **Flat league** | 5/6 | Jackson Chourio 15.7%, Junior Caminero 14.2%, William Contreras 13.6%, Jake Bauers 13.0%, Yandy Díaz 11.9%, Christian Yelich 11.7% |
| **KNN comps** | 6/6 | Jackson Chourio 16.0%, Junior Caminero 14.7%, Jake Bauers 14.2%, William Contreras 13.8%, Christian Yelich 11.9%, Jonathan Aranda 11.8% |
| **SVR** | 5/6 | Jackson Chourio 16.3%, Junior Caminero 16.0%, Jake Bauers 14.0%, William Contreras 13.7%, Yandy Díaz 12.1%, Jonathan Aranda 11.4% |
| **Random forest** | 5/6 | Jackson Chourio 16.2%, Junior Caminero 15.2%, Jake Bauers 14.2%, William Contreras 13.7%, Yandy Díaz 11.9%, Christian Yelich 11.7% |
| **Core (KNN+SVR)** | 6/6 | Jackson Chourio 16.0%, Junior Caminero 14.7%, Jake Bauers 14.2%, William Contreras 13.8%, Christian Yelich 11.9%, Jonathan Aranda 11.8% |
| **Forest blend** | 6/6 | Jackson Chourio 16.0%, Junior Caminero 14.7%, Jake Bauers 14.2%, William Contreras 13.8%, Christian Yelich 11.9%, Jonathan Aranda 11.8% |
| **Linear regression** | 6/6 | Jackson Chourio 16.2%, Junior Caminero 14.9%, Jake Bauers 14.0%, William Contreras 13.7%, Christian Yelich 11.8%, Jonathan Aranda 11.7% |
| **Logistic regression** | 5/6 | Jackson Chourio 16.2%, Junior Caminero 15.3%, Jake Bauers 14.0%, William Contreras 13.7%, Yandy Díaz 11.6%, Jonathan Aranda 11.5% |
| **Neural net (torch)** | 5/6 | Jackson Chourio 16.2%, Junior Caminero 15.3%, Jake Bauers 14.1%, William Contreras 13.8%, Yandy Díaz 11.7%, Jonathan Aranda 11.6% |
| **XGBoost** | 5/6 | Jackson Chourio 16.2%, Junior Caminero 15.5%, Jake Bauers 14.3%, William Contreras 13.7%, Yandy Díaz 12.0%, Christian Yelich 11.6% |

## Hitter by hitter

The top 40 bats by the served probability.

| | HITTER | | SERVED (FULL SEASON) | FLAT LEAGUE | KNN COMPS | SVR | RANDOM FOREST | CORE (KNN+SVR) | FOREST BLEND | LINEAR REGRESSION | LOGISTIC REGRESSION | NEURAL NET (TO |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | **Jackson Chourio** | MIL | **15.4%** | 15.7% | 16.0% | 16.3% | 16.2% | 16.0% | 16.0% | 16.2% | 16.2% | |
| 2 | **Junior Caminero** | TB | **14.9%** | 14.2% | 14.7% | 16.0% | 15.2% | 14.7% | 14.7% | 14.9% | 15.3% | |
| 3 | **Jake Bauers** | MIL | **14.1%** | 13.0% | 14.2% | 14.0% | 14.2% | 14.2% | 14.2% | 14.0% | 14.0% | |
| 4 | **William Contreras** | MIL | **13.8%** | 13.6% | 13.8% | 13.7% | 13.7% | 13.8% | 13.8% | 13.7% | 13.7% | |
| 5 | **Jonathan Aranda** | TB | **11.7%** | 11.1% | 11.8% | 11.4% | 11.4% | 11.8% | 11.8% | 11.7% | 11.5% | |
| 6 | **Christian Yelich** | MIL | **11.6%** | 11.7% | 11.9% | 11.3% | 11.7% | 11.9% | 11.9% | 11.8% | 11.4% | |
| 7 | **Yandy Díaz** | TB | **11.5%** | 11.9% | 11.6% | 12.1% | 11.9% | 11.6% | 11.6% | 11.7% | 11.6% | |
| 8 | **Brice Turang** | MIL | **10.4%** | 10.6% | 10.1% | 10.3% | 10.0% | 10.1% | 10.1% | 9.7% | 9.7% | |
| 9 | **Ryan Vilade** | TB | **9.5%** | 8.9% | 9.4% | 9.1% | 9.4% | 9.4% | 9.4% | 9.1% | 8.9% | |
| 10 | **Garrett Mitchell** | MIL | **9.5%** | 9.1% | 9.4% | 8.8% | 9.1% | 9.4% | 9.4% | 9.1% | 9.1% | |
| 11 | **Jonny DeLuca** | TB | **7.3%** | 8.4% | 7.4% | 7.3% | 7.0% | 7.4% | 7.4% | 7.4% | 7.7% | |
| 12 | **Cooper Pratt** | MIL | **7.1%** | 8.5% | 7.0% | 6.5% | 6.3% | 7.0% | 7.0% | 6.0% | 6.1% | |
| 13 | **Liam Hicks** | TB | **6.8%** | 7.8% | 6.8% | 7.5% | 7.4% | 6.8% | 6.8% | 7.2% | 7.1% | |
| 14 | **David Hamilton** | MIL | **6.5%** | 7.7% | 6.3% | 6.2% | 5.8% | 6.3% | 6.3% | 6.4% | 6.3% | |
| 15 | **Sal Frelick** | MIL | **6.5%** | 8.1% | 6.8% | 6.1% | 6.0% | 6.8% | 6.8% | 6.5% | 6.6% | |
| 16 | **Ben Williamson** | TB | **4.5%** | 7.6% | 5.1% | 4.5% | 4.5% | 5.1% | 5.1% | 3.2% | 4.9% | |
| 17 | **Taylor Walls** | TB | **3.7%** | 4.8% | 3.5% | 2.6% | 2.7% | 3.5% | 3.5% | 3.4% | 3.5% | |
| 18 | **Chandler Simpson** | TB | **2.0%** | 4.5% | 2.6% | 1.2% | 1.3% | 2.6% | 2.6% | 1.1% | 2.3% | |

slate

Per-game probabilities: each prior's per-PA rate run through this hitter's own plate appearances, park, weather, bullpen and times-through-the-order terms. Only the prior changes across the columns — every other term is held fixed, so a left-to-right spread is the prior disagreeing about the hitter, not the context. Shading grows with the probability.

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